Emily Herbert

herbert.emilyanne.jobs@gmail.com
emilyaherbert.com
github.com/emilyaherbert

Engineer specializing in AI systems, distributed inference, and developer tooling under real-world constraints.

Education

2021 **MS in Computer Science**, *The University of Massachusetts Amherst*, Amherst, MA.

2018 **BS in Computer Science**, *Trinity University*, San Antonio, TX.

2025 **Art & Design Intensive**, *Atelier de Sevres*, Paris, France.

Relevant Experience

Oct 2025 – Apr 2026 **TAHO (early-stage startup)**.
Product Engineer - Rust
- Deployed large-scale models (Qwen3) on commodity hardware under memory/latency constraints, validated up to 32B parameters.
- Built distributed inference components using streaming-first decomposition to enable execution without full model materialization.
- Investigated limitations in inference frameworks (especially ONNX in Rust) and synthesized constraints to guide system design.
- Built tooling (graph visualization, TUI) used in team workflows to debug and inspect model execution.
- Integrated PyTorch, ONNX, and onnxruntime to bridge model formats and execution runtimes.

Aug 2021 - May 2023 **Fuel Labs (early-stage startup)**.
Compiler Engineer - Rust
- Contributed to the Sway smart contract language and compiler, used by a large developer ecosystem (61k+ GitHub stars).
- Implemented core language features (match expressions, traits, generics) informed by programming language research.
- Translated developer feedback into language and tooling improvements.
- Delivered technical talks and demos to onboard developers and explain system design.

May 2021 – Aug 2021 **Google**, *Madison, WI*.
Software Engineering Intern - C++
- Implemented load balancing in a high-performance networking library using RMA-based RPC mechanisms.
- Integrated low-level systems for efficient communication with network hardware.

June 2018 - May 2021 **Northeastern University, University of Massachusetts Amherst**, *Boston, MA, Amherst, MA*.
PhD Researcher - Rust, TypeScript, Python
- Built a serverless function accelerator compiling traced JavaScript into Rust to improve latency and throughput.
- Led development across compiler, runtime, networking, and storage layers in a $0 \rightarrow 1$ system.
- Implemented logging and debugging systems to diagnose non-deterministic behavior under limited observability.
- Built and trained neural network models using TensorFlow/Keras with data preprocessing and evaluation pipelines.

June 2017 – Aug 2017 **National Aeronautics and Space Administration (NASA)**, *Langley, VA*.
Software Engineering Intern - C++
- Contributed to GPS-based geofencing system for autonomous drones across heterogeneous platforms.
- Tested and validated system behavior using simulated hardware environments.

Selected Talks & Publications

2023 Blockchain Language Design panel. *Layer 2 Day at EthDenver.* [recording]

2022 Developing Smart Contracts in Sway talk. *Layer2Amsterdam.* [slides]

Sway: A Rust-based Smart Contract Language talk. *EthDenver.* [slides]

2021 Emily Herbert and Arjun Guha. A Language-based Serverless Function Accelerator. [preprint, repo]

2020 Wang Cen, Emily A Herbert, and Peter J Haas. NIM: Modeling and Generation of Simulation Inputs via Generative Neural Networks. *Winter Simulation Conference.* [paper]. **Best Contributed Theoretical Paper Finalist.**

Skills

Core Languages, Rust (primary), TypeScript, Python, Haskell, Scala, Sway .

AI/ML, PyTorch, ONNX, onnxruntime, TensorFlow/Keras, NumPy, Hugging Face .

Systems, Distributed systems, runtime design, IR design, low-latency systems, Protobuf, async (Tokio), rkyv .

Tools, Docker, Kubernetes, AWS (EC2, Lambda, S3), GCP, React, TUI (ratatui), CLI (clap), debugging, VSCode .